

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Dynamic equilibria, feedback & tipping points - Set 1

Mark scheme

AQA spec 3.6.1; Sustainability; textbook pp. 370-372

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs	1
01.2	• Impact depends on total population x per capita impact • With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added • Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates	3
01.3	• Urban areas are supported by resources brought in from elsewhere; when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage • Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need	2
01.4	• Climate: warming increases evaporation, cloud and albedo; higher CO₂ increases photosynthesis and sequestration; hydrological cycle: more evaporation gives more precipitation; populations: density-dependent factors return numbers to carrying capacity • In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) • Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	

Question	Marking guidance	Marks
02.1	- Negative feedback responds to any change by re-establishing the previous equilibrium - Only large unusual events overwhelm the feedback so a new equilibrium forms - Example: methane hydrate release at the end of the Permian causing climate change and mass extinction	3
02.2	- In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change - Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate - Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather - Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source	4
02.3	<i>Any 3 from:</i> - Melting permafrost releasing methane - Reduced albedo as ice / snow melt - Methane hydrate release - Forest and peat fires - Faster soil decomposition - Ocean acidification reducing sequestration - Cirrus cloud formation	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
03.1	- Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people - Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so	2
03.2	- Total impact = population size x the individual (per capita) impact of each person - Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact	2
03.3	<i>Any 1 from:</i> - Slash and burn agriculture - Fishing - Discharge of sewage / biodegradable pollutants	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
04.1	- Negative - Positive - Negative - Positive	4
04.2	- The response to a change acts to re-establish / return the system to the previous equilibrium - So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback	2
04.3	- eg the possible release of methane hydrate at the end of the Permian era - Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium	2

Question	Marking guidance	Marks
04.4	• Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) • Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
05.1	• The new opportunity is overwhelmingly attractive (economic benefit) • Lack of knowledge / understanding of the impacts • The impacts occur elsewhere or much later, so are not evident to those who benefit	3
05.2	• Cultural belief that resources should be exploited to satisfy needs only • Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes	2
05.3	• Resources are brought in from other areas • If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing	2
05.4	• Easier access to energy has made many new activities possible • The scale of activities has greatly increased • Toxic wastes are produced that cannot be decontaminated by natural processes	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	